

OpenStax Calculus Volume 3

Section 4.4 Exercises: Tangent Planes and Linear Approximations

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For the following exercises, find a unit normal vector to the surface at the indicated point.

1. $z = f(x, y) = x^3, (2, -1, 8)$
2. $\ln\left(\frac{x}{y-z}\right) = 0$ when $x = y = 1$

For the following exercises, as a useful review for techniques used in this section, find a normal vector and a tangent vector at point P .

3. $x^2 + xy + y^2 = 3, P(-1, -1)$
4. $(x^2 + y^2)^2 = 9(x^2 - y^2), P(\sqrt{2}, 1)$
5. $xy^2 - 2x^2 + y + 5x = 6, P(4, 2)$
6. $2x^3 - x^2y^2 = 3x - y - 7, P(1, -2)$
7. $ze^{x^2-y^2} - 3 = 0, P(2, 2, 3)$

For the following exercises, find the equation for the tangent plane to the surface at the indicated point. (Hint: Solve for z in terms of x and y .)

8. $-8x - 3y - 7z = -19, P(1, -1, 2)$
9. $z = -9x^2 - 3y^2, P(2, 1, -39)$
10. $x^2 + 10xyz + y^2 + 8z^2 = 0, P(-1, -1, -1)$
11. $z = \ln(10x^2 + 2y^2 + 1), P(0, 0, 0)$
12. $z = e^{7x^2+4y^2}, P(0, 0, 1)$
13. $xy + yz + zx = 11, P(1, 2, 3)$
14. $x^2 + 4y^2 = z^2, P(3, 2, 5)$
15. $x^3 + y^3 = 3xyz, P(1, 2, \frac{3}{2})$
16. $z = axy, P(1, \frac{1}{a}, 1)$
17. $z = \sin x + \sin y + \sin(x + y), P(0, 0, 0)$

18. $z = h(x, y) = \ln \sqrt{x^2 + y^2}, P(3, 4)$

19. $z = x^2 - 2xy + y^2, P(1, 2, 1)$

For the following exercises, find parametric equations for the normal line to the surface at the indicated point. (Recall that to find the equation of a line in space, you need a point on the line, $P_0(x_0, y_0, z_0)$, and a vector $n = \langle a, b, c \rangle$ that is parallel to the line. Then the equation of the line is $x - x_0 = at, y - y_0 = bt, z - z_0 = ct$.)

20. $-3x + 9y + 4z = -4, P(1, -1, 2)$

21. $z = 5x^2 - 2y^2, P(2, 1, 18)$

22. $x^2 - 8xyz + y^2 + 6z^2 = 0, P(1, 1, 1)$

23. $z = \ln(3x^2 + 7y^2 + 1), P(0, 0, 0)$

24. $z = e^{4x^2 + 6y^2}, P(0, 0, 1)$

25. $z = x^2 - 2xy + y^2$ at point $P(1, 2, 1)$

For the following exercises, use the figure shown here.

[Figure: A surface in the xyz -plane is marked as $z = f(x, y)$. This surface has a tangent plane at (x_0, y_0, z_0) , with the corresponding point (x_0, y_0) marked on the xy -plane. Also marked on the xy -plane is the point $(x_0 + \Delta x, y_0 + \Delta y)$. From this point, a line is drawn to the surface and three points are marked. The first point is C , which is $(x_0 + \Delta x, y_0 + \Delta y, z_0)$, then there is B , which is on the tangent plane, and then there is A , which is on the surface. The distance between B and C is marked $df(x_0, y_0)$.]

26. The length of line segment AC is equal to what mathematical expression?

27. The length of line segment BC is equal to what mathematical expression?

28. Using the figure, explain what the length of line segment AB represents.

For the following exercises, complete each task.

29. Show that $f(x, y) = e^{xy}x$ is differentiable at point $(1, 0)$.

30. Find the total differential of the function $w = e^y \cos(x) + z^2$.

31. Show that $f(x, y) = x^2 + 3y$ is differentiable at every point. In other words, show that $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y) = f_x \Delta x + f_y \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$, where both ϵ_1 and ϵ_2 approach zero as $(\Delta x, \Delta y)$ approaches $(0, 0)$.

32. Find the total differential of the function $z = \frac{xy}{y+x}$ where x changes from 10 to 10.5 and y changes from 15 to 13.

33. Let $z = f(x, y) = xe^y$. Compute Δz from $P(1, 2)$ to $Q(1.05, 2.1)$ and then find the approximate change in z from point P to point Q . Recall $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$, and dz and Δz are approximately equal.

34. The volume of a right circular cylinder is given by $V(r, h) = \pi r^2 h$. Find the differential dV . Interpret the formula geometrically.

35. See the preceding problem. Use differentials to estimate the volume of aluminum in an enclosed aluminum can with diameter 8.0 cm and height 12 cm if the aluminum is 0.04 cm thick.
36. Use the differential dz to approximate the change in $z = \sqrt{4 - x^2 - y^2}$ as (x, y) moves from point $(1, 1)$ to point $(1.01, 0.97)$. Compare this approximation with the actual change in the function.
37. Let $z = f(x, y) = x^2 + 3xy - y^2$. Find the exact change in the function and the approximate change in the function as x changes from 2.00 to 2.05 and y changes from 3.00 to 2.96.
38. The centripetal acceleration of a particle moving in a circle is given by $a(r, v) = \frac{v^2}{r}$, where v is the velocity and r is the radius of the circle. Approximate the maximum percent error in measuring the acceleration resulting from errors of 3% in v and 2% in r . (Recall that the percentage error is the ratio of the amount of error over the original amount. So, in this case, the percentage error in a is given by $\frac{da}{a}$.)
39. The radius r and height h of a right circular cylinder are measured with possible errors of 4% and 5%, respectively. Approximate the maximum possible percentage error in measuring the volume (Recall that the percentage error is the ratio of the amount of error over the original amount. So, in this case, the percentage error in V is given by $\frac{dV}{V}$.)
40. The base radius and height of a right circular cone are measured as 10 in. and 25 in., respectively, with a possible error in measurement of as much as 0.1 in. each. Use differentials to estimate the maximum error in the calculated volume of the cone.
41. The electrical resistance R produced by wiring resistors R_1 and R_2 in parallel can be calculated from the formula $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$. If R_1 and R_2 are measured to be 7Ω and 6Ω , respectively, and if these measurements are accurate to within 0.05Ω , estimate the maximum possible error in computing R . (The symbol Ω represents an ohm, the unit of electrical resistance.)
42. The area of an ellipse with axes of length $2a$ and $2b$ is given by the formula $A = \pi ab$. Approximate the percent change in the area when a increases by 2% and b increases by 1.5%.
43. The period T of a simple pendulum with small oscillations is calculated from the formula $T = 2\pi\sqrt{\frac{L}{g}}$, where L is the length of the pendulum and g is the acceleration resulting from gravity. Suppose that L and g have errors of, at most, 0.5% and 0.1%, respectively. Use differentials to approximate the maximum percentage error in the calculated value of T .
44. Electrical power P is given by $P = \frac{V^2}{R}$, where V is the voltage and R is the resistance. Approximate the maximum percentage error in calculating power if 120 V is applied to a $2000 - \Omega$ resistor and the possible percent errors in measuring V and R are 3% and 4%, respectively.

For the following exercises, find the linear approximation of each function at the indicated point.

45. $f(x, y) = x\sqrt{y}$, $P(1, 4)$
46. $f(x, y) = e^x \cos y$, $P(0, 0)$
47. $f(x, y) = \arctan(x + 2y)$, $P(1, 0)$
48. $f(x, y) = \sqrt{20 - x^2 - 7y^2}$, $P(2, 1)$
49. $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, $P(3, 2, 6)$

50. [T] Find an equation of the tangent plane to the surface $f(x, y) = x^2 + y^2$ at point $(1, 2, 5)$, and graph the surface and the tangent plane at the point.
51. [T] Find the equation for the tangent plane to the surface at the indicated point, and graph the surface and the tangent plane: $z = \ln(10x^2 + 2y^2 + 1)$, $P(0, 0, 0)$.
52. [T] Find an equation of the tangent plane to the surface $z = f(x, y) = \sin(x + y^2)$ at point $(\frac{\pi}{4}, 0, \frac{\sqrt{2}}{2})$, and graph the surface and the tangent plane.